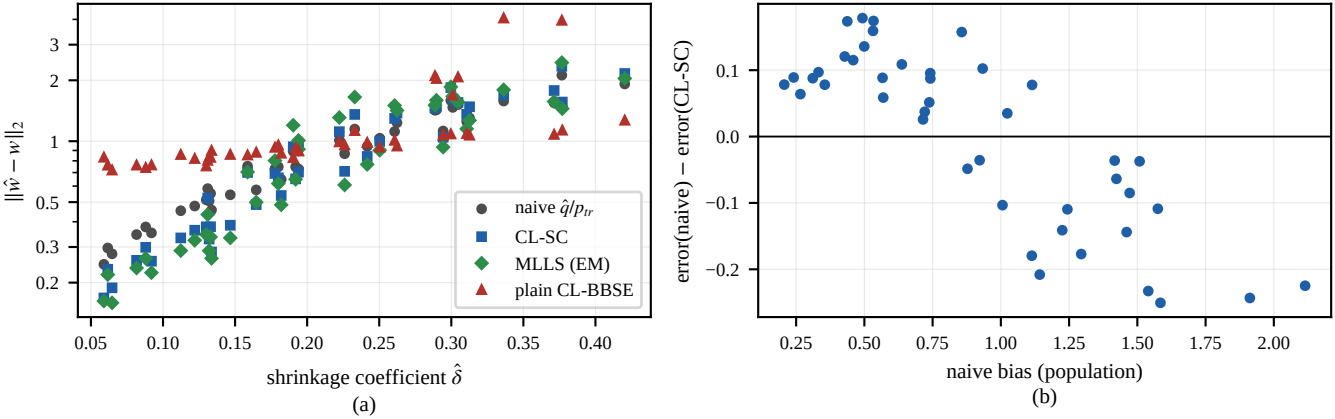
$\|\hat{w} - w\|_2$
3
2
1
0.5
0.3
0.2
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
shrinkage coefficient $\hat{\delta}$
naive $\hat{q}/p_{tr}$
CL-SC
MLLS (EM)
plain CL-BBSE
error(naive) − error(CL-SC)
0.1
0.0
−0.1
−0.2
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
naive bias (population)

(a) (b)